

Master's Thesis Project - Data Matrix Decoding

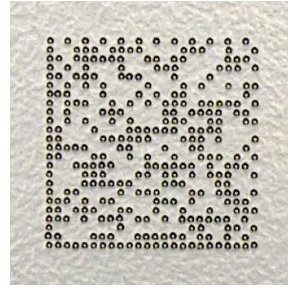
April 15, 2025

Background

In industrial engineering settings, Data Matrix codes are often marked directly onto machined components to serialize them. The serialization process is carried out with either laser marking or dot peen marking techniques, which can be seen in Figure 1. Although most widely available Data Matrix scanners work in ideal conditions, they often fail when used from mobile phones in industrial settings.



(a) Laser marked.



(b) Dot-peen marked.

Figure 1: Example showing the difference between laser and dot-peen marked data matrix codes.

Decoding failure can occur due to poor quality cameras, the Data Matrix codes being physically small, light reflections on metal surfaces, and different kinds of blur. The majority of scanning software are also not designed to recognize dot peen marked Data Matrix codes, a commonly used method of marking components due to being cheaper than laser marking.

Table 1 shows the typical decoding process for decoding a DMC from image input.

Step	Purpose	Traditional Approach
Image Preprocessing	To reduce noise and errors, and increase efficiency in later steps.	The image may be grayscaled, or binarized with a specified threshold.
Localization	To find an approximate area of where the DMC is located in the image. Further reduces noise and increases efficiency in later steps.	Algorithmic approach usually based on locating DMC finder pattern ("L" shape) via e.g. edge detection.
Geometric Alignment	To find the orientation of the DMC. Map to grid-wise representation for later decoding. Standardizes input for later steps.	Algorithmic approach based on edge detection, finder pattern analysis, corner detection, perspective correction.
Symbol Decoding	Extracting DMC encoded data (bit matrix).	The grid produced in prev step is iterated over, and the contrast between pixels belonging to DMC and background are used to convert DMC into encoded bits. Most common standard also uses Reed-Solomon error correction.
Data Extraction	Final decoding step to reveal encoded info.	Standard DMC decoding calculation.

Table 1: Common steps traditionally used for decoding data matrix codes.

Today AI and machine learning methods have advanced enough that they can be utilized to replace some of the steps and produce better results. Here are some example of methods that have been tried in previous work:

- Better binarization can be done with image segmentation techniques, but this is very computationally expensive to do accurately.
- Object detection to produce bounding box of DMC location. Real-time models like YOLO can do object detection faster and more accurately than traditional methods. Furthermore, YOLO models that produce oriented bounding boxes have potential to remove noise even more.
- Keypoint detection to produce coordinates of corners of DMC. With the corners of the DMC located, the DMC can be rectified, increasing decode rate. This is however tricky to learn for keypoint detection models.
- For decoding dot-peen marked DMCs, a complicated process involving UNet can be used to locate some of the dots (same approach as braille detection) to gather template candidates, followed by template matching to locate all dots within the image. Once the x,y coordinates of every dot is found algorithmic approaches similar to traditional methods can be used for decoding.

Previous research into this area has been done by one of our student assistants and their papers and code can be found on GitHub. [Research Project \(7.5 ECTS\)](#) where YOLO and U-Net were used to preprocess images. [Master's Thesis \(30 ECTS\)](#) where YOLO and a new method for decoding is being used. Note that the Master's project is ongoing until the 2nd of June.

Your Project

Your project can either aim to adjust and improve upon previously used methods, or aim to research new methods for comparison.

The methods used should either be generalized to work both with laser and dot-peen marked codes, or multiple methods specialized on laser and dot-peen marked codes should be used together. Methods for dot-peen marked codes should be robust to misplacement errors that commonly occur from the dot-peen marking process.

A baseline decoder using a traditional algorithmic approach ([libdmtx](#)) should be used for comparing successful decode rate and runtime to methods using AI.

An image dataset provided by MAN ES can be used for training and comparing the performance of different methods. As of writing the dataset contains 207 images, but more images will be provided by MAN ES during the course of the project. There is also the opportunity to collect more images manually at the factory located in Sydhavn.